Active Lesion Localization in LITS CT Scans Using Deep Reinforcement Learning

GitHub Repository: https://github.com/Dannyism11/Lession_Detection_in_RL
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Abstract

We study liver tumor localization in abdominal CT as an active sequential decision problem. Instead of predicting a bounding box in one forward pass, the agent begins from a broad region, repeatedly transforms the box, and terminates when the localized region is sufficiently aligned with a lesion. LiTS17 CT volumes and tumor masks are used to construct 2D axial localization episodes from 3D lesion annotations. We evaluate three stages under a common action space and shared VGG16-fc6 representation: a Deep Q-Network (DQN) baseline, a stabilized Proximal Policy Optimization (PPO) localizer, and multi-start confidence-based reporting. The DQN baseline reached validation mean best IoU of 0.203 and test mean best IoU of 0.129. The stabilized PPO localizer raised validation mean best IoU to 0.464, with 72% of validation cases reaching best IoU ≥ 0.25 and 58% reaching best IoU ≥ 0.5, but terminal stopping remained weak because 86% of validation episodes timed out. Multi-start confidence reporting raised validation mean best IoU to 0.528 and test mean best IoU to 0.234, with test success-best@0.5 increasing from 4% to 16% relative to the first PPO run. These results support the hypothesis that policy-gradient training improves lesion-directed navigation, while the remaining challenge is reliable conversion of high-IoU intermediate states into final reports.

1. Introduction

Accurate lesion localization is a central subproblem in computer-aided diagnosis, treatment planning, and downstream segmentation. Liver tumors in abdominal CT are especially difficult because lesions can be small, irregular, low contrast, and sparse relative to the full image. A dense detector or segmenter must learn to separate a small foreground target from a very large anatomical background, while a two-stage pipeline must first crop a reliable region of interest before the fine model can operate effectively. The LiTS benchmark was designed around this difficulty: its CT volumes contain tumors with varied size and appearance, and the published benchmark reports that tumor segmentation remains considerably harder than liver segmentation (Bilic et al., 2023).

We reformulate lesion localization as active visual search. The agent observes the current image crop and the history of previous actions, selects one of eight geometric transformations or a terminal trigger action, and receives reward according to whether the action improves overlap with the lesion. This framing follows the active object localization idea of Caicedo and Lazebnik (Caicedo & Lazebnik, 2015), where localization is treated as a Markov decision process rather than as exhaustive sliding-window search or one-shot box regression. The medical motivation is also aligned with anatomical landmark localization work, where deep reinforcement learning is used to learn efficient search paths instead of scanning all spatial hypotheses (Ghesu et al., 2016; 2019; Abdullah Al & Yun, 2020).

The goal is not full tumor segmentation. Instead, we isolate the localization stage: given tumor masks during training and evaluation, 3D lesion annotations are converted into 2D axial bounding-box records, and an agent is trained to move from the whole slice toward the lesion. This scope is narrower than the full LiTS challenge, but it isolates a decision-making component that is relevant to two-stage segmentation and detection pipelines. If an RL agent can learn where to crop, a downstream segmentation model can operate on a more balanced foreground-background region.

The experimental progression contains three stages:

1. Baseline: a Caicedo-style DQN localizer using frozen VGG16-fc6 features, action history, normalized box geometry, replay memory, target-network bootstrapping, guided exploration, and

a trigger action.

2. Stabilized PPO actor-critic: a model using the same observation representation but replacing off-policy Q-learning with clipped policy optimization, generalized advantage estimation, behavior cloning from a greedy oracle, action masking, curriculum learning, entropy annealing, and stricter reward shaping.

3. Multi-start confidence reporting: a deployment-time reporting improvement over the same PPO policy, using deterministic multi-start evaluation and stop-confidence scoring to recover high-quality intermediate boxes when the learned trigger policy fails.

The central empirical finding is that the stabilized PPO actor-critic substantially improved localization potential, as measured by the best IoU reached during an episode, but did not solve the stopping problem. Multi-start confidence reporting directly targeted that gap by reporting the most confident intermediate state instead of always trusting the terminal state.

2. Related Work

2.1. Active Object Localization

Caicedo and Lazebnik (Caicedo & Lazebnik, 2015) proposed one of the most direct precedents for this work: a class-specific visual agent that starts with a large image region, applies discrete bounding-box transformations, and terminates with a trigger action. Their formulation is attractive because it turns localization into a sequence of low-dimensional control decisions. Rather than evaluating thousands of proposals, the agent processes a small number of attended regions. The baseline implementation follows this design closely: it uses eight movement or shape actions plus a trigger action, a VGG16 feature vector for the current crop, and a history vector representing the previous actions.

Sequential localization was later extended to multiple objects and proposal generation. Jie et al. (Jie et al., 2016) introduced tree-structured reinforcement learning so that multiple search policies can discover several objects with diverse trajectories. This supports the broader motivation for sequential search: history matters, and the best next crop depends on the path already taken. Our formulation remains single-target per episode, but the record builder creates one episode per target lesion, which is a practical simplification for multi-lesion CT volumes.

2.2. Deep Reinforcement Learning for Medical Images

Medical image localization differs from natural-image localization because the target often occupies a tiny fraction of the image, the visual texture is subtle, and spatial context is anatomical rather than semantic. Ghesu et al. (Ghesu et al., 2016) framed anatomical landmark detection as a behavior-learning task in which an agent learns search paths instead of relying on exhaustive scanning. Their later multi-scale work (Ghesu et al., 2019) showed that RL-based navigation can improve both accuracy and speed for 3D CT landmark detection. Abdullah Al and Yun (Abdullah Al & Yun, 2020) further argued for actor-critic policy learning in medical landmark localization, motivated by faster behavior learning and better convergence than pure value-based search in large state spaces.

The stabilized PPO actor-critic is motivated by this line of work. The baseline DQN is a natural starting point because value-based deep RL is simple for discrete actions. However, medical image localization has sparse successful terminal events and many visually similar states. In such settings, actor-critic methods can reduce variance through a learned value baseline while directly optimizing the action policy. PPO is particularly suitable because its clipped objective constrains large policy updates without the implementation complexity of second-order trust-region methods (Schulman et al., 2017).

Recent surveys emphasize the same practical issue: DRL combines representation learning with sequential control, but sample efficiency, reward design, and stability remain central difficulties (Coronato et al., 2020; Wang et al., 2024; Hua et al., 2022). The implementation addresses these difficulties through frozen visual features, reward shaping, behavior cloning from a greedy oracle, action masks, target KL stopping, value clipping, and a curriculum over lesion diameter.

2.3. Medical Lesion Data and Two-Stage Systems

The experiments use the LiTS17 liver tumor segmentation data. LiTS contains 131 training CT volumes and was designed to evaluate liver and tumor segmentation under varied tumor sizes and contrast appearances (Bilic et al., 2023). We convert segmentation masks into box-based localization episodes, so the labels are used to define an active localization task rather than to train a dense segmentation model.

The decision to solve localization rather than segmentation is also supported by two-stage medical image pipelines. Segmentation frameworks such as nnU-Net (Isensee et al., 2018) are strong when the target re-

Table 1. Dataset split and 2D record construction. PPO runs use all lesion records for metadata construction and cap training/evaluation records during training.

Split	Volumes	3D lesions	2D records
Train	89	624	624
Validation	14	132	132
Test	15	128	128

gion is presented at a useful scale, but small-object segmentation still suffers from class imbalance and background dominance. Recent RL-based collaborative localization and segmentation work (Xu et al., 2026) explicitly argues that localization and segmentation should exchange information rather than be optimized as disconnected stages. This work does not implement the segmentation stage, but it develops the RL localization component that such a pipeline would need.

3. Dataset and Problem Formulation

3.1. Data Construction

The experiments use the Kaggle LiTS-style directory structure:

CT_Vol/CT_Vol, CT_Mask/CT_Mask.

The total configured volume count is 131. The segmentation masks are parsed for tumor label 2. Connected 3D tumor components with at least 10 voxels are retained, and each component is converted into a 3D bounding box. Any bounding-box dimension smaller than 4 voxels is padded to a minimum dimension of 4 so that very thin components still form usable localization targets.

The data split is volume-level rather than slice-level. This avoids leakage from adjacent slices of the same patient volume into different splits. Table 1 summarizes the split used in the final runs: 89 training volumes, 14 validation volumes, and 15 test volumes, corresponding to 624, 132, and 128 3D lesions respectively. Each lesion becomes one 2D axial localization record by selecting the lesion center slice and projecting the 3D tumor bounding box onto the axial plane.

All CT intensities are clipped to the Hounsfield window $[-200, 300]$ and resized to 224×224 crops before visual encoding. This window is consistent with the liver-lesion setting because it suppresses extreme air and bone values while retaining soft-tissue contrast. The crop is repeated to three channels and normalized with ImageNet mean and standard deviation before passing through VGG16.

3.2. MDP Definition

Each 2D record defines an episodic Markov decision process

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma).$$

The state s_t contains the current image crop, a 90-dimensional action-history vector, and an 8-dimensional normalized box-geometry vector:

$$[x_1/W, y_1/H, x_2/W, y_2/H, c_x/W, c_y/H, w_b/W, h_b/H].$$

The history length is 10 and there are 9 actions, so the flattened action history has $10 \times 9 = 90$ entries. The image crop is encoded by a frozen VGG16-fc6 extractor into a 4096-dimensional feature vector. The policy or Q-network therefore receives a 4194-dimensional non-image input after visual encoding.

The action set is:

$$\begin{aligned}\mathcal{A} = \{&\text{right}, \text{left}, \text{down}, \text{up}, \text{bigger}, \\ &\text{smaller}, \text{fatter}, \text{taller}, \text{trigger}\}.\end{aligned}$$

Movement and shape actions transform the current box by a factor $\alpha = 0.20$ relative to the current width or height. The trigger action terminates the episode. Episodes begin from the full axial slice for the baseline and the stabilized PPO localizer, and are capped at 40 steps. The IoU between the current box b_t and a lesion box g is

$$\mathrm{IoU}(b_t, g) = \frac{|b_t \cap g|}{|b_t \cup g|}.$$

3.3. Evaluation Metrics

Evaluation reports both terminal and trajectory metrics. This distinction is essential. The final IoU is the IoU of the reported box, while the best IoU is the maximum IoU reached anywhere in the trajectory. Success-final@0.5 measures whether the final report is a strict detection success, whereas success-best@0.5 measures whether the agent ever reached a good localization state. Trigger rate, successful trigger rate, bad trigger rate, timeout rate, and mean steps explain whether failure is caused by poor navigation or poor stopping.

This distinction prevents overclaiming. A high best-IoU but low final-IoU means the search policy can find the lesion but the terminal decision loses it. That is precisely what occurs in the stabilized PPO localizer and directly motivates multi-start confidence reporting.

4. Methodology

4.1. Baseline: DQN Active Localizer

The baseline follows the Caicedo active localization pattern. A frozen VGG16-fc6 encoder maps the current crop to a 4096-dimensional vector. The Q-network concatenates this vector with box features and action history, then applies two 1024-unit fully connected layers with ReLU activations and dropout 0.5. The final layer outputs Q-values for the 9 discrete actions. The model has 5,354,505 trainable Q-network parameters. The VGG16 encoder is pretrained and frozen, so the baseline only learns the decision head.

The Bellman target for a transition (s_t, a_t, r_t, s_{t+1}) is

$$y_t = r_t + \gamma(1 - d_t) \max_{a'} Q_{\bar{\theta}}(s_{t+1}, a'),$$

with $\gamma = 0.95$ and a target network $Q_{\bar{\theta}}$. The model minimizes a Huber loss between $Q_\theta(s_t, a_t)$ and y_t. Replay memory size is 50,000; the optimizer is Adam with learning rate 10^{-4}; batch size is 32; learning starts after 500 transitions; and the target network updates every 500 steps.

The baseline reward is deliberately simple: movement receives $+1$ when IoU improves, -1 when IoU decreases, and 0 when unchanged. Triggering at IoU ≥ 0.5 gives $+3$, while triggering below 0.5 gives -3. This reward is theoretically aligned with the Bellman objective because each local step estimates whether the action increases expected future overlap. However, it is also a weak signal for terminal calibration: a near-threshold bad trigger and a very poor trigger both receive the same terminal penalty, while timeout is not as strongly discouraged as in the PPO runs.

Several engineering decisions make the DQN baseline more stable:

- epsilon-greedy exploration anneals from 1.0 to 0.05;
- guided exploration samples positive IoU-improving actions and demonstrates the trigger when IoU is already high;
- trigger-positive replay oversamples successful trigger transitions;
- reversal suppression prevents immediate undo actions;
- no-op masking removes geometric actions that do not change the clipped box.

Figure 1 shows the baseline architecture and action preview.

4.2. Stabilized PPO Localizer

The stabilized PPO localizer keeps the environment, action space, VGG16-fc6 representation, history, and box features, but replaces DQN with an actor-critic PPO policy. This is the main methodological change. DQN learns action values off-policy from replay, while PPO directly optimizes a stochastic policy using on-policy rollouts. The PPO clipped surrogate objective is

$$\bar{r}_t(\theta) = \mathrm{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon),$$
$$L^{\mathrm{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \bar{r}_t(\theta)\hat{A}_t \right) \right].$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t|s_t)}.$$

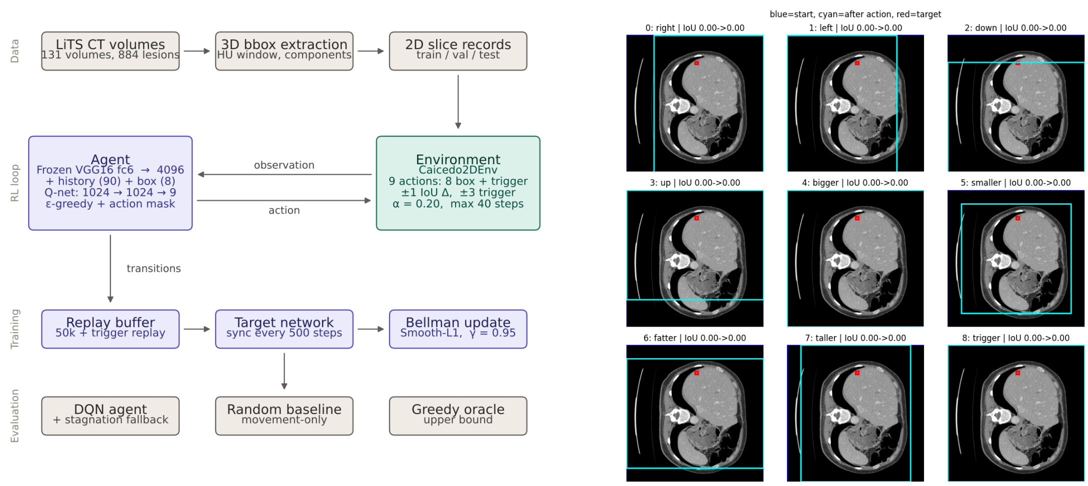
The clipping coefficient is 0.10. Advantages are computed with generalized advantage estimation using $\gamma = 0.95$ and $\lambda = 0.95$, then normalized per minibatch. The value loss is clipped with coefficient 0.20, the value coefficient is 0.25, gradient norms are clipped at 0.5, and optimization stops early if the approximate KL exceeds the target 0.012. These choices directly address a known PPO risk: repeated minibatch updates on the same rollout can destabilize the policy if the new policy moves too far from the sampled policy (Schulman et al., 2017).

The actor-critic network has the same input dimensionality as the DQN head. A shared two-layer 1024-unit torso feeds a 9-action policy head and a scalar value head. The policy has 5,355,530 trainable parameters. The VGG16 encoder has 117,479,232 frozen parameters and zero trainable encoder parameters in the final run, so the stabilized PPO localizer is not explained by a larger visual backbone. It is explained by the training objective, rollout handling, reward design, and curriculum.

The stabilized PPO localizer adds five important stabilizers.

Behavior cloning from the oracle. Before PPO, the policy is trained for 5 epochs on 500 greedy-oracle episodes. The oracle greedily selects IoU-improving actions and triggers at the evaluation threshold. On a 64-record validation sample, the oracle reaches mean best IoU 0.539, success-best@0.25 of 1.0, success-best@0.5 of 1.0, trigger rate 1.0, and mean 14.91 steps. This check proves that the action space can solve representative episodes before learning begins.

Action masking. Invalid no-op actions and direct reversals are masked by setting their logits to a very negative value before sampling or evaluating the categorical policy. This improves exploration quality be-



(a) Baseline DQN architecture.

(b) Baseline action preview from one environment state.

Figure 1. Baseline model and action space. The DQN encodes the current crop with frozen VGG16-fc6 features and predicts Q-values for eight box transformations plus trigger. The action preview shows how each discrete action changes the current box before learning.

cause PPO does not waste probability mass on actions that are impossible or immediately undo the previous step.

Strict trigger shaping. The trigger reward is made threshold-centered and continuous: -5.0 at IoU 0.0, +5.0 at IoU 0.5, and +7.0 at IoU 1.0. Timeout receives -5.0. This aligns the reward with the reported IoU@0.5 success metric more strongly than the baseline trigger reward.

Auxiliary trigger loss. States with IoU $\geq$ 0.5 are encouraged to place margin on the trigger logit, while states with IoU $\leq$ 0.25 penalize trigger confidence. This auxiliary loss is weighted by 0.50. It directly targets the difficult terminal decision rather than relying only on delayed rollout returns.

Curriculum learning. Training begins with larger lesions, transitions to medium lesions, then uses all lesions: 20 mm for 30% of steps, 10 mm for 30%, and all records for 40%. This is theoretically justified by curriculum learning: the agent first learns coarse geometric behavior on easier targets before facing smaller, more ambiguous lesions.

4.3. Multi-Start Confidence Reporting

Multi-start confidence reporting targets a specific failure exposed by the stabilized PPO localizer: the policy often reaches a high-IoU box but does not report it as the final box. In the final evaluation table, stabilized PPO validation mean best IoU is 0.464, but mean final IoU is only 0.109 and timeout rate is 0.86. The agent is therefore searching better than it is stopping.

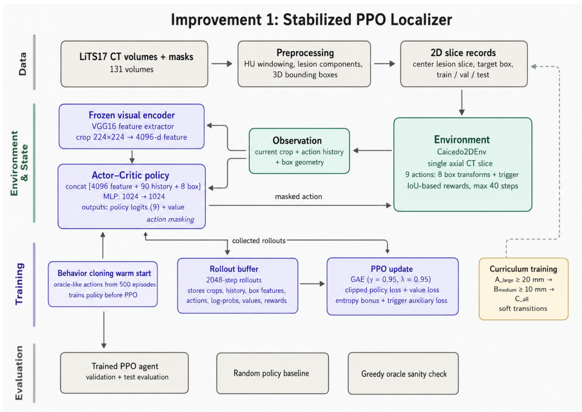
Multi-start confidence reporting changes the reporting protocol while keeping the same trained PPO localizer structure. It computes a stop-confidence score at each frame using the policy logits and value prediction. The default score is the stop margin

$$m_t = z_t^{\text{trigger}} - \max_{a<8} z_t^a,$$

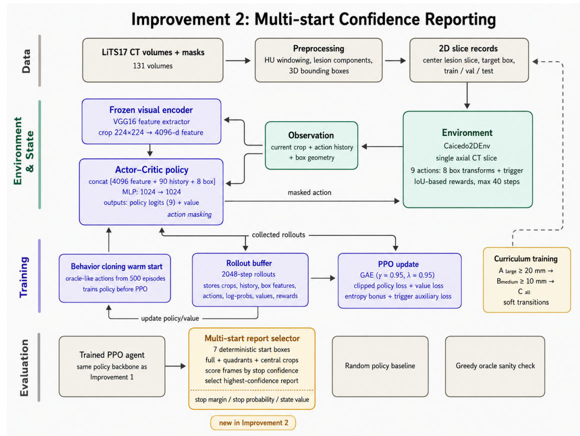
where z_t are masked policy logits. A large positive margin means the policy prefers stopping over every movement action. If an episode times out, the system reports the finite frame with the highest stop-confidence score instead of blindly reporting the last frame. This converts the policy's internal stopping preference into a deployment-time confidence signal.

Multi-start confidence reporting also evaluates seven deterministic starting boxes: full image, four quadrants, a 10% padded crop, and a central 40% crop. Candidate reported boxes must occur after at least one step and occupy at most 75% of the image area. The final reported frame is selected as the highest-confidence candidate across all starts. This is a conservative multi-start design: it does not require random test-time sampling, and the starts are deterministic, reproducible, and cheap relative to full dense search.

Multi-start confidence reporting is theoretically motivated by partial observability and local search. Starting from the full image can make small lesions nearly



(a) Stabilized PPO localizer.



(b) Multi-start confidence reporting.

Figure 2. Architecture diagrams for the two improvement stages. The stabilized PPO localizer replaces DQN with an actor-critic PPO policy over the same visual and geometric state. Multi-start confidence reporting retains the trained PPO policy and adds deterministic multi-start evaluation with stop-confidence report selection.

invisible after resizing; starting from a quadrant or central crop can increase local resolution and change the policy's confidence landscape. The reported metric confirms that this helps on test: mean multistart gain is +0.0245 IoU, and test best-IoU improves from 0.134 in the stabilized PPO localizer to 0.234 with multi-start confidence reporting.

5. Experimental Setup

5.1. Shared Settings

All three runs use seed 42, maximum episode length 40, action scale $\alpha = 0.20$, minimum box size 8 pixels, image size 224×224, VGG16 ImageNet weights, 4096-dimensional fc6 features, 10-step action history, 8 normalized box features, and REWARD_MODE=any. Training records are capped at 300 and validation evaluation during training is capped at 100 records. The PPO final test evaluation uses the first 100 test records because MAX_VAL_RECORDS=100 is reused for test capping; the baseline final test table reports all 128 test records. For fair baseline-to-improvement claims, validation numbers are therefore the primary common comparison. The stabilized PPO localizer and multi-start confidence reporting use identical test size and can be compared directly on test.

5.2. Diagnostic and Qualitative Evidence

The experiments contain three types of diagnostic evidence beyond final metric tables. First, the greedy oracle verifies that the action space can reach useful boxes. Second, training curves show that validation best IoU increases during training even when terminal behavior remains unstable. Third, qualitative rollouts show the policy repeatedly shrinking and shifting boxes toward the tumor, often reaching a high-IoU intermediate state before later degradation or timeout.

Figure 3 summarizes the training diagnostics retained for analysis. The baseline curve shows noisy validation behavior across only 8 epochs. The PPO curve contains rollout-level policy loss, value loss, entropy, KL, rollout reward, best-IoU, and validation success diagnostics, reflecting a much more controlled policy-optimization procedure.

6. Results

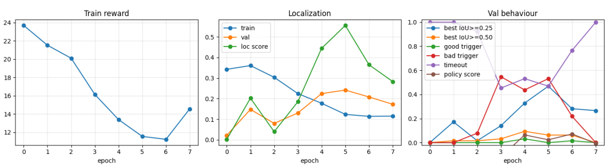
6.1. Final Quantitative Results

Table 3 gives the central comparison. The strongest validation trend is best-IoU improvement: baseline 0.203, stabilized PPO 0.464, and multi-start confidence reporting 0.528. Success-best@0.5 follows the same progression: 5%, 58%, and 65%. This is the clearest evidence that PPO learns better navigation than the DQN baseline.

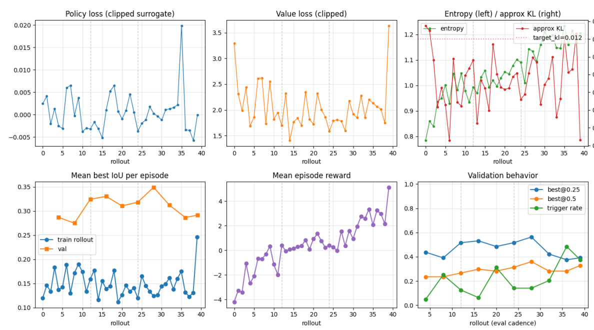
Final-IoU metrics tell a more nuanced story. The stabilized PPO localizer improves best-IoU but has low final-IoU because most episodes time out and the terminal box is not necessarily the best box found. Multi-start confidence reporting explicitly changes the report to confidence-selected intermediate states, raising validation final-IoU from 0.109 to 0.158 and test final-IoU from 0.044 to 0.082 relative to the stabilized PPO lo-

Table 2. Key training configuration. The stabilized PPO localizer and multi-start confidence reporting use the same policy size as the baseline decision head but change the training algorithm and reporting protocol.

Component	Baseline DQN	Stabilized PPO	Multi-start reporting
Visual encoder	Frozen VGG16-fc6	Frozen VGG16-fc6	Frozen VGG16-fc6
Decision parameters	5,354,505	5,355,530	Same PPO policy
Optimizer	Adam, 10^{-4}	Adam, 3×10^{-5}	No extra training
Training budget	8 epochs	80,000 steps	Uses best PPO checkpoint
Batching	Replay, batch 32	Rollout 2048, minibatch 64	7 deterministic starts
RL objective	Bellman Q-loss	PPO clipped actor-critic	Stop-confidence selection
Discount	0.95	0.95	Same policy
Trigger threshold	IoU 0.5	IoU 0.5	IoU 0.5
Main additions	Guided exploration, replay	BC, GAE, masks, curriculum	Multi-start, stop margin



(a) Baseline DQN training diagnostics.



(b) PPO training diagnostics.

Figure 3. Training diagnostics. The DQN baseline improves unevenly across a short epoch schedule, while PPO tracks optimization terms, rollout reward, best-IoU, and validation metrics over 80k environment steps.

calizer. Test success-best@0.5 also rises from 4% to 16%.

The validation oracle provides an upper diagnostic reference for the discrete action space rather than a learned-model result. In the baseline run, the oracle validation sample reaches mean best IoU 0.543 with 98% success-best@0.25 and 94% success-best@0.5. In the PPO runs, the 64-record oracle sample reaches mean best IoU 0.539, success-best@0.25 of 1.0, and success-best@0.5 of 1.0. Multi-start confidence reporting validation mean best IoU of 0.528 is close to this oracle-level localization potential, which suggests that the remaining limitation is trigger/report calibration rather than the available transformation actions.

6.2. Multi-Start Reporting Diagnostics

Table 4 reports diagnostic metrics for the reporting protocol. The raw terminal final-IoU values match the stabilized PPO localizer because the underlying policy is the same. The reported final-IoU changes after single-start confidence fallback and multi-start selection. On test, mean single-start final-IoU is 0.057 and the multi-start gain is +0.0245, producing the final reported test IoU of 0.082. On validation, multi-start gain is negative on average even though best-IoU improves. This means multi-start confidence reporting is helpful but not perfectly calibrated; it sometimes selects a worse frame despite finding better boxes somewhere in the trajectory.

6.3. Qualitative Rollout

Figure 4 uses the baseline validation rollout and the IoU trace from the same rollout. This example is useful because it shows the intended active-search behavior even in the baseline: the box is repeatedly transformed toward the lesion, reaches a best IoU of 0.832 at step 34, and reports at step 40 with IoU 0.473. The gap between the best intermediate state and the reported state motivates the later multi-start confidence reporting protocol.

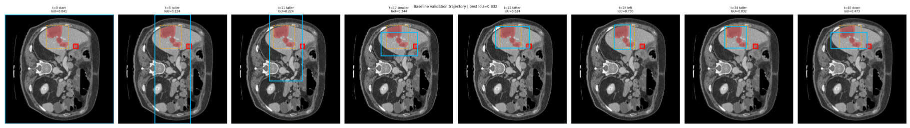
7. Discussion

7.1. Why Stabilized PPO Improved Localization

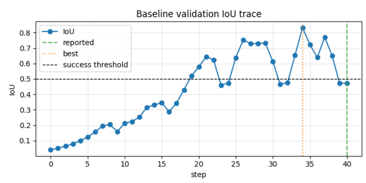
The strongest empirical claim is that the stabilized PPO localizer improved localization potential. On vali-

Table 3. Final evaluation metrics. Baseline test uses 128 test records; PPO test rows use 100 records. Validation rows use 100 records for all three stages and are the fairest progression comparison.

Stage	Split	n	Final IoU	Best IoU	Final@0.25	Final@0.5	Best@0.25	Best@0.5	Timeout
Baseline DQN	Val	100	0.134	0.203	0.210	0.010	0.320	0.050	0.760
Stabilized PPO	Val	100	0.109	0.464	0.170	0.050	0.720	0.580	0.860
Multi-start reporting	Val	100	0.158	0.528	0.230	0.040	0.780	0.650	0.860
Baseline DQN	Test	128	0.057	0.129	0.055	0.008	0.227	0.047	0.875
Stabilized PPO	Test	100	0.044	0.134	0.060	0.020	0.220	0.040	0.780
Multi-start reporting	Test	100	0.082	0.234	0.120	0.050	0.350	0.160	0.780



(a) Validation rollout.



(b) IoU trace for the same rollout.

Figure 4. Validation rollout and corresponding IoU trace. The same trajectory reaches a high-IoU intermediate localization before reporting a lower-IoU terminal box, exposing the search-versus-stopping gap addressed by multi-start confidence reporting.

Table 4. Multi-start reporting diagnostics. These numbers show how much of the gain comes from confidence fallback and multi-start selection.

Split	Raw final	Single-start	Multi gain	Improved	Worse
Val	0.109	0.184	-0.026	0.25	0.35
Test	0.044	0.057	0.025	0.13	0.20

dation, mean best-IoU increased from 0.203 to 0.464 after the stabilized PPO localizer. This is too large to attribute to a different backbone or a larger model: both stages use frozen VGG16-fc6 features and a roughly 5.35M-parameter decision head. The difference is the training procedure.

Several mechanisms explain the stabilized PPO localizer. First, PPO optimizes the policy directly instead of relying on max-Q bootstrapping over a replay distribution. In sparse terminal tasks, Q-values for rarely successful trigger states can be noisy, and off-policy replay may preserve outdated behavior. PPO collects on-policy trajectories and updates the policy under a clipped ratio, making changes more conservative. Second, GAE uses the value function to trade off bias and variance in advantage estimates, which is useful when rewards are local but success depends on a multi-step path. Third, behavior cloning initializes the policy near oracle geometric behavior, reducing the burden on exploration. Fourth, action masking removes invalid and counterproductive choices before PPO samples them. Fifth, the curriculum increases task difficulty gradually, which is appropriate because small tumors produce weaker visual and reward signals.

7.2. Why Final IoU Lagged Behind Best IoU

The results show that localization and stopping are different skills. The stabilized PPO localizer often moves to good boxes but times out 86% of validation episodes and 78% of test episodes. The trigger action is hard because it is both a classification decision and a control decision: the model must infer that the current region is good enough, compare stopping to all possible future transformations, and commit to termination. A small error in trigger calibration can convert a successful search into a poor final report.

The strict trigger reward and auxiliary trigger loss improved the theoretical alignment between training and evaluation, but they did not fully solve stopping. One reason is class imbalance at the state level. During a

40-step rollout, many more states are below the trigger threshold than above it, and high-IoU states may be brief. Another reason is perceptual ambiguity: a cropped lesion box near IoU 0.5 can look visually similar to a box at IoU 0.35, especially after resizing and VGG feature extraction. The final result therefore supports a more precise conclusion: PPO learned a useful search policy, but the terminal policy needs stronger supervision or a separate calibrated report head.

7.3. Why Multi-Start Reporting Helped

Multi-start confidence reporting is justified because it attacks the observed failure mode without retraining the entire model. If the learned policy internally assigns high trigger logit margin to a good intermediate state, then this margin can be used as a confidence score even when the action selected later causes timeout. Reporting the best stop-confidence frame is similar in spirit to using model confidence for detection ranking, except the candidates are generated along an RL trajectory.

Multi-start confidence reporting adds another benefit through multi-start evaluation. The original full-image start can make small targets hard to see after resizing. Starting from quadrants or a central crop changes the initial scale and can expose the lesion earlier. The test results show this helps: multi-start confidence reporting raises test best-IoU from 0.134 to 0.234 and success-best@0.5 from 4% to 16%. However, the validation diagnostic also shows that confidence is not perfectly calibrated, because the multi-start average gain is negative on validation. A future version should learn a supervised report-quality head or use validation-calibrated thresholds instead of relying only on stop-margin.

7.4. Limitations

The main limitation is that the experiments evaluate 2D center-slice localization, not full 3D lesion localization. The 3D masks are used to generate targets, but the agent acts only on axial 2D boxes. This simplifies the action space and makes the experiments feasible, but it ignores through-plane lesion extent and can miss clinically relevant 3D context.

Another limitation is trigger reliability. Even after strict reward shaping and auxiliary trigger supervision, timeout remains high. The system should therefore not be interpreted as a complete detector. It is better interpreted as evidence that the search policy can produce useful candidate boxes, and that the report stage requires additional calibration.

A further limitation is data efficiency. The PPO run uses 80,000 environment steps and behavior cloning from 500 oracle episodes, but the effective training set is capped at 300 records. More records, cross-validation over volume splits, and repeated seeds would be necessary before making robust claims about generalization.

8. Conclusion

We presented an active lesion localization pipeline for CT using deep reinforcement learning. The baseline DQN established a Caicedo-style control formulation with frozen VGG16 features, discrete box transformations, action history, replay learning, and trigger-based termination. The stabilized PPO actor-critic replaced DQN and added behavior cloning, action masking, strict trigger rewards, auxiliary trigger supervision, and curriculum learning. This produced a clear validation localization gain: mean best-IoU rose from 0.203 to 0.464, and validation success-best@0.5 rose from 5% to 58%. Multi-start confidence reporting recognized that the remaining issue was reporting rather than navigation. By using stop-confidence fallback and seven deterministic starts, it raised validation mean best-IoU to 0.528 and test mean best-IoU to 0.234.

Overall, the stabilized PPO localizer made the agent substantially better at reaching lesion boxes, while multi-start confidence reporting improved outputs when the trigger policy failed. Future work should move from 2D to 3D actions, train a calibrated report-quality head, compare against supervised bounding-box detectors, and connect the localized crops to a downstream segmentation network.

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